

Homework 2 – STAT 6611

Due Friday, October 2 by midnight (submit in gradescope under ‘Assignments’)

1. Suppose that $\underline{y}$ is an $n \times 1$ observable random vector that follows Model II.

- (a) If $\underline{\lambda}'_1 \underline{\beta}$ and $\underline{\lambda}'_2 \underline{\beta}$ are both estimable, is $\underline{\lambda}'_1 \underline{\beta} + \underline{\lambda}'_2 \underline{\beta}$ necessarily estimable? Explain your answer.
- (b) If $\underline{\lambda}'_1 \underline{\beta}$ and $\underline{\lambda}'_2 \underline{\beta}$ are both nonestimable, is $\underline{\lambda}'_1 \underline{\beta} + \underline{\lambda}'_2 \underline{\beta}$ necessarily nonestimable? Explain your answer.
- (c) Can $\underline{\lambda}'_1 \underline{\beta} + \underline{\lambda}'_2 \underline{\beta}$ be estimable if $\underline{\lambda}'_1 \underline{\beta}$ is estimable and $\underline{\lambda}'_2 \underline{\beta}$ is nonestimable? Explain your answer.

2. Take the model to be Model VII and suppose that

$$\underline{y} = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix} \quad \text{and} \quad X = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

- (a) Find the orthogonal projections of $\underline{y}$ on $\mathcal{C}(X)$ and on $\mathcal{C}^\perp(X)$.
- (b) Determine $\text{rank}(X'X)$.
- (c) Use the following method and find a generalized inverse of $X'X$.

Let A be an $n \times m$ matrix of rank $r \geq 1$. Let B be a nonsingular $r \times r$ submatrix of A . For matrix A , replace the entries of B by those of $(B^{-1})'$, and replace all other entries of A by 0. Then the transpose of the resulting matrix is a generalized inverse of A .

- (d) Compute a solution $\hat{\underline{\beta}}$ to the normal equations.
- (e) Check that the vectors $X\hat{\underline{\beta}}$ and $\underline{y} - X\hat{\underline{\beta}}$ agree with your results in part (a).

3. For Model VII, partition $\underline{y}$, X , and $\underline{\epsilon}$ correspondingly as follows:

$$\underline{y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}, \quad X = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}, \quad \text{and} \quad \underline{\epsilon} = \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \end{bmatrix}.$$

Suppose that the row spaces of X_1 and X_2 are essentially disjoint, that is, $\mathcal{R}(X_1) \cap \mathcal{R}(X_2) = \{0\}$. Show that, if $\underline{\lambda}' \in \mathcal{R}(X_1)$, then the least squares estimate of $\underline{\lambda}' \underline{\beta}$ based on the model $y_1 = X_1 \underline{\beta} + \epsilon_1$ is the same as the least squares estimate of $\underline{\lambda}' \underline{\beta}$ based on the model $\underline{y} = X \underline{\beta} + \underline{\epsilon}$.

Hint: Under the full model, the LSE is $\underline{a}' X' \underline{y} = \underline{a}' X'_1 y_1 + \underline{a}' X'_2 y_2$ and, under the partial model, the LSE is $\underline{b}' X'_1 y_1$ where here $X'_1 X_1 \underline{a} + X'_2 X_2 \underline{a} = X' X \underline{a} = \underline{\lambda} = X'_1 X_1 \underline{b}$ (why?); in the last equation, move terms involving X'_2 and X'_1 to different sides.

4. Show that, under Model II, $\underline{a}' \underline{y}$ is a BLUE of its expectation if and only if $\underline{a} \in \mathcal{C}(X)$.

5. (a) Show that, under Model II, the problem of determining the BLUE $\underline{a}' \underline{y}$ of an estimable function $\underline{\lambda}' \underline{\beta}$ is equivalent to that of minimizing $\underline{a}' \underline{a}$ subject to the constraints $\underline{a}' X = \underline{\lambda}'$.
- (b) The Lagrangian function associated with this minimization problem is

$$L(\underline{a}, \underline{\rho}) = \underline{a}' \underline{a} + 2 \underline{\rho}' (X' \underline{a} - \underline{\lambda}).$$

Use the vector calculus to differentiate the Lagrangian function with respect to $\underline{a}$ and the Lagrangian multipliers, and set these derivatives equal to zero.

- (c) Show that the resulting equations have a solution if and only if the conjugate normal equations are consistent.
- (d) Solve your equations for the minimizing value of $\underline{a}$.